

NHS Digital Hospital Agent

2-Month Bulk Task Plan — Full End-to-End AI Agent Platform

Domains: Python · Data Science · Data Analytics · Full Stack · AI/ML · Gen AI

Structure: 4 Bi-Weekly Sprints (Weeks 1–2, 3–4, 5–6, 7–8)

Scope: a full end-to-end agent covering patient-facing services (triage, appointment booking, Q&A; chatbot), clinical staff support (notes summarization, discharge letters, records lookup), and hospital ops/admin (scheduling, resource allocation, analytics). Tasks are listed as a flat backlog per domain per sprint — assign owners as needed for your team size.

Contents

Sprint 1 (Weeks 1–2): Foundation & Discovery

Environment setup, data audit, wireframes, model research

Sprint 2 (Weeks 3–4): Core Feature Build

Booking APIs, dashboards, triage model v1, chatbot v1

Sprint 3 (Weeks 5–6): Intelligence Layer & Integration

Predictive models, RAG chatbot, admin panel, MLOps

Sprint 4 (Weeks 7–8): Testing, Compliance & Deployment

UAT, safety cases, security testing, go-live

NHS Compliance & Governance Checklist

DTAC, DSPT, DCB0129/0160, GDPR, interoperability standards

Consolidated Domain Task Summary

Full task count per domain across all sprints

Sprint 1 — Weeks 1–2: Foundation & Discovery

Goal: environments ready, requirements gathered, baselines defined across every domain.

PYTHON

- Set up repo structure, coding standards, and branching strategy
- Scaffold backend service using FastAPI (or Django REST)
- Set up authentication module (OAuth2 / NHS CIS2 login stub)
- Configure logging, monitoring, and error-tracking (e.g. Sentry)
- Build CI/CD pipeline skeleton (lint, test, build, deploy stages)
- Connect to a test/sandbox FHIR server for EHR data exchange
- Set up environment configs for dev/staging/prod

DATA SCIENCE

- Audit available datasets (synthetic/sample hospital data) for quality and gaps
- Exploratory data analysis on appointments, no-shows, and admissions data
- Define core KPIs: no-show rate, average wait time, bed occupancy, readmission rate
- Build initial data cleaning and preprocessing pipeline
- Draft feature list for downstream prediction models
- Document data dictionary and assumptions

DATA ANALYTICS

- Gather reporting requirements from clinical and admin stakeholders
- Set up BI tool connection (Power BI / Looker / Metabase) to sample data
- Build first-cut bed occupancy and wait-time dashboard (mock data)
- Define standard report templates (daily ops summary, weekly trends)
- Set up data pipeline from source systems into the analytics layer

FULL STACK

- Create UX wireframes: patient portal, staff dashboard, chatbot widget
- Set up frontend repo (React/Next.js) with component library
- Build design system (colors, typography, accessible components per NHS Digital Service Manual)
- Build basic auth pages (login, registration, password reset)
- Connect frontend skeleton to backend API stub

AI/ML

- Research triage severity classification approaches (rule-based vs ML)
- Define model requirements for a symptom-checker / triage assistant
- Collect and label a sample dataset for triage severity (synthetic/anonymized)
- Build a baseline rule-based triage classifier for comparison
- Draft model evaluation criteria and success metrics

GEN AI

- Select and provision LLM provider/API for the project
- Set up a prompt-engineering sandbox and versioned prompt repository
- Design conversational flow for the patient-facing chatbot (intake, FAQ, escalation)
- Draft system prompts for clinical note summarization use case
- Draft initial safety/guardrail requirements (PII handling, escalation to human, refusal behavior)

Sprint 2 — Weeks 3–4: Core Feature Build

Goal: first working versions of booking, dashboards, triage model, and chatbot.

PYTHON

- Implement appointment booking API (create/reschedule/cancel)
- Build patient record CRUD against mock EHR/FHIR resources
- Implement scheduling engine logic (slot allocation, conflict checks)
- Integrate NHS number validation (Modulus 11 check algorithm)
- Set up background job queue (Celery/RQ) for reminders and notifications
- Write unit tests for booking and records modules

DATA SCIENCE

- Build no-show prediction model v1 (baseline classifier)
- Feature engineering pipeline for appointment/patient history data
- Evaluate model with precision/recall/F1, not just accuracy
- Draft initial resource allocation logic (bed/staff demand estimation)
- Document model assumptions and limitations

DATA ANALYTICS

- Build wait-time and bed occupancy dashboard v1 (live data pipeline)
- Build department-level KPI report (A&E, outpatients, wards)
- Automate daily/weekly report generation and distribution
- Implement data quality checks and alerting for pipeline failures

FULL STACK

- Build patient portal: booking flow, profile, chatbot widget embed
- Build staff dashboard: patient queue, priority flags, alerts
- Integrate frontend with booking and records APIs
- Implement responsive design across mobile/tablet/desktop
- Accessibility pass targeting WCAG 2.1 AA (required for NHS digital services)

AI/ML

- Develop triage severity classification model v1 (ML-based)
- Integrate triage output into the booking flow as a priority queue signal
- Set up train/validation/test split with stratification by severity class
- Produce a model explainability report (feature importance, example cases)

GEN AI

- Build patient-facing conversational agent v1 (symptom intake + FAQ answering)
- Build clinical note summarization prototype for staff review
- Implement retrieval-augmented generation (RAG) over NHS policy/FAQ knowledge base
- Add hallucination guardrails and source citation for generated answers
- Add conversation logging with PII redaction for audit purposes

Sprint 3 — Weeks 5–6: Intelligence Layer & Integration

Goal: predictive models and generative features connected end-to-end, with monitoring and safety checks in place.

PYTHON

- Full FHIR-compliant data exchange integration (patient, encounter, observation resources)
- Build discharge summary generation pipeline (data assembly for Gen AI drafting)
- Build notification service (SMS/email reminders and alerts)
- Add error handling, retries, and circuit breakers for external integrations
- Harden API security: rate limiting, input validation, audit logging

DATA SCIENCE

- Refine no-show model with additional features (weather, distance, history)
- Build readmission risk model
- Build staff/bed demand forecasting model (time-series)
- Design an A/B testing framework for future model rollouts
- Cross-validate models against held-out time periods

DATA ANALYTICS

- Build executive-level hospital-wide KPI dashboard
- Integrate DS model outputs (risk scores, forecasts) into visualizations
- Set up automated anomaly-detection alerts (e.g. sudden occupancy spike)
- Write data governance documentation (lineage, retention, access control)

FULL STACK

- Surface AI/ML outputs in UI (triage flags, readmission risk indicators)
- Build an admin panel for model monitoring and override controls
- Implement multi-language support for patient-facing screens
- Performance optimization pass (bundle size, API latency, caching)

AI/ML

- Deploy triage and readmission models to a staging environment
- Set up model monitoring for data/prediction drift
- Run a fairness/bias audit across age, gender, and ethnicity groups
- Set up an MLOps pipeline: versioning, experiment tracking, retraining triggers

GEN AI

- Expand chatbot to handle multi-turn clinical queries with context memory
- Build a discharge-letter drafting assistant for clinician review and edit
- Implement a human-in-the-loop review workflow for all AI-generated clinical content
- Run red-team testing for prompt injection, hallucination, and unsafe advice scenarios

Sprint 4 — Weeks 7–8: Testing, Compliance & Deployment

Goal: validated, safety-signed-off platform ready for go-live.

PYTHON

- End-to-end integration testing across booking, records, and notification services
- Load and performance testing (simulate peak A&E;/booking traffic)
- Prepare for third-party security penetration testing
- Finalize CI/CD pipeline for production deployment with rollback support
- Set up backup and disaster recovery procedures

DATA SCIENCE

- Final validation of all models against clinical safety acceptance criteria
- Produce model cards (purpose, data, performance, limitations) for each model
- Present model performance and risk assessment to the clinical governance board

DATA ANALYTICS

- Run user acceptance testing (UAT) on dashboards with clinical/admin stakeholders
- Set up automated alerting and reporting cadence for go-live
- Validate historical data migration into production analytics store

FULL STACK

- UAT sessions with clinical staff and patient representatives
- Full accessibility audit (WCAG 2.1 AA) with remediation
- Cross-browser and cross-device testing
- Bug fixing and polish pass
- Production deployment and smoke testing

AI/ML

- Complete clinical safety case documentation (aligned to DCB0129 principles)
- Final bias/fairness sign-off before production release
- Deploy models to production with monitoring dashboards live
- Document and test the rollback plan for model failures

GEN AI

- Final safety and guardrail testing: jailbreak resistance, PII leakage checks
- Finalize human-oversight protocol for all patient- and clinician-facing AI output
- Prepare staff training materials for using the AI assistant safely
- Prepare a go-live support plan and escalation path for AI-related incidents

NHS Compliance & Governance Checklist

Any NHS-facing digital agent needs to satisfy specific governance requirements alongside the feature build. These should be tracked as ongoing workstreams across all four sprints, not left to the end.

Requirement	What it covers
DTAC (Digital Technology Assessment Criteria)	Baseline assessment for clinical safety, data protection, technical security, interoperability, and usability/accessibility before NHS adoption.
DSPT (Data Security and Protection Toolkit)	Annual self-assessment against NHS data security standards; required for any organization handling NHS patient data.
DCB0129 / DCB0160	Clinical risk management standards for health IT systems; requires a named Clinical Safety Officer and a documented safety case.
UK GDPR / Data Protection Act 2018	Lawful basis for processing patient data, data minimization, right to explanation for automated decisions.
Interoperability standards	HL7 FHIR UK Core for data exchange, SNOMED CT for clinical coding, NHS number as the patient identifier.
Accessibility	WCAG 2.1 AA compliance is a legal requirement for NHS public-facing digital services.
Human oversight	Any AI-generated clinical content (summaries, letters, triage flags) requires clinician review before acting on it.

Consolidated Domain Task Summary

Domain	Sprint 1	Sprint 2	Sprint 3	Sprint 4	Total
Python	7	6	5	5	23
Data Science	6	5	5	3	19
Data Analytics	5	4	4	3	16
Full Stack	5	5	4	5	19
AI/ML	5	4	4	4	17
Gen AI	5	5	4	4	18
All Domains	33	29	26	24	112

Total backlog: 112 tasks across 6 domains over 4 bi-weekly sprints (8 weeks). Assign owners per your actual team size — tasks within a sprint can run in parallel across people in the same domain.